

朱新源

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教育背景

耶鲁大学	计算机科学	理学硕士	2025. 8-2026. 5
香港中文大学（深圳）	计算机工程	工学学士	2021. 7-2025. 7
GPA: 3.87/4.00 (1/78) IELTS: 7.5 计算机技能: Python (Tensorflow, Pytorch), C++, C, Java, Verilog, SQL, Arduino IDE, SolidWorks, 安卓开发 荣誉奖项: 2025 届大学杰出毕业生 (1%), 香港中文大学（深圳）优秀学生奖 (0.1%), 本科生 A 等学业奖学金 (8w 人民币), 理工学院院长荣誉录, 思廷书院院长嘉许奖, 德勤最佳校园大使			
慕尼黑工业大学	计算机科学与技术	交换生	2024. 3-2024. 9

实习经历

博世（中国）	AI 大模型实习生	2025. 5-2025. 7
- 主导开发团队代码上线流程优化, 完成测试代码生成大模型开发全流程, 基于内部代码库 (4000+条代码-Gtest 代码对), 采用 LoRA 微调优化博世内部大模型, 提升代码测试覆盖率 12% (79%→91%), 支持团队日均 300+次调用。 - 优化智驾需求对接流程, 基于 QLoRA 微调本地大模型, 开发端到端语义理解系统, 实现客户需求文档的意图识别与结构化信息抽取, 将生活化语言转化为工程师端标准技术文档, 相比人工流程缩短开发周期 30+%。		
OPPO 广东移动通信有限公司	软件开发实习生	2024. 8-2024. 10
- 开发项目进度自动告警机器人, 基于 Python 开发多线程参数爬虫自动监控网站参数, 捕获项目异常并触发通信平台警报, 集成检测数据网页与 OPPO 公司内部通讯软件 TeamTalk API 实现智能告警, 实现 100%异常拦截。 - 参与 OPPO 新版本应用商店开发上线, 负责 Android 用户界面迭代与功能优化, 使用 Kotlin 在 Android Studio 编写并执行 23 个版本测试用例, 成功拦截 4 个 P0 级关键 BUG, 确保新版本按期完成开发、测试与上线。		
中信证券	机器学习实习生	2024. 2-2024. 3
- 为提升股票市场价格走势预测准确度, 采用 XGBoost 机器学习算法模型融合 MA (30/60/120)+MACD (12, 26, 9) 等技术指标, 对沪深 300 指数历史交易数据进行特征工程与非线性关系建模, 提升价格走势预测能力, 实现预测准确率 70+%。 - 为提升投资收益, 结合 XGBoost 与 SHAP 值解释提取股价主要影响因子并动态调整权重, 提升模型可解释性。		
香港应用科技研究院	机器学习实习生	2023. 8-2023. 9
- 支持硅晶板性能优化决策, 构建多项式回归+随机森林混合模型, 处理温度-性能曲线, 准确拟合晶片性能曲线。 - 完成电池寿命 RUL 预测算法, 使用 Pandas 处理 10 万+行传感器 CSV 数据, 训练多项式回归模型, 结合可视化模块实现电池 RUL 预测, 支持企业电池健康管理与维护决策。		

科研经历

香港中文大学（深圳）毕业设计	作者	2024. 9-2025. 5
- 开发基于卷积神经网络的声音信号处理模型, 融合 MFCC/声谱图特征实现 92%弦乐演奏技巧识别准确率。 - 设计多模态特征融合架构 (Mel-Spectrogram + 时域特征), 引入空洞卷积与 Attention 机制强化长序列建模, 进而提升弦乐及弹拨乐短时与长时演奏特征识别性能, 相关成果发表于论文 (1)。		
莱布尼兹超级计算中心（慕尼黑工业大学）	强化学习组组长	2024. 4-2024. 8
- 调整机器学习的 Actor-Critic 强化学习模型, 调整学习率以减少 57%仿真-现实差异。 - 比较机械臂关节信号以及仿真环境输出, 并根据差异位对 Mujoco 仿真器添加高斯扰动, 覆盖质量/阻尼/惯量等 12 个维度, 调整目标策略平滑系数编写程序, 动态调整 PID 控制器的 Kp/Ki/Kd, 实现差异进一步降低。总结海报。		
深圳市人工智能与机器人研究院	本科生科研助理	2022. 10-2024. 3
- 为提升帆船定位准确度, 应用卡尔曼滤波以及多项式回归构建轨迹减扰模型, 成功提升 83%帆船位置捕捉准确度。 - 开发基于自回归模型的帆船环境自适应算法, 结合新型减阻增稳装置, 实现帆船航行动态调整, 并发表相关论文 (2)。		

论文发表

- X. Y. Zhu, C. Leung (2025). CNNs in Musical Performance and Arrangement: Recognizing and Managing Bowed Instrument Techniques Across Cultures, 2025 AIMEDIA, Venice, Italy.
- X. Y. Zhu^[1], C. Liang^[2], H. H. Qian* (2023). Design and Implementation of a Novel Adaptive Multihull Sailboat with Lifiable Side Hulls*, 2023 IEEE International Conference on Robotics and Biomimetics, Koh Samui, Thailand.

Xinyuan Zhu

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EDUCATION

Yale University	Master of Computer Science (Class of 2026)	New Haven, USA
The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)		Shenzhen, China
B.Eng. Electronic Information Engineering (Stream: Computer Engineering)		Sept 2021 - June 2025
- Major GPA: 3.87/4.00 (Rank: 1/78). IELTS: 7.5 - Computer Skills: Python (Tensorflow, Pytorch), C++, C, Java, Verilog, SQL, SolidWorks, Android Studio - Honors & Awards: Outstanding Graduate of Class 2025 (top 1%), Excellent Student Award (top 0.1%), Academic Performance Scholarship (8w RMB), Dean's List, Master's List, Best Ambassador of Deloitte		
Technical University of Munich	Exchange Student	Mar 2024 - Aug 2024
		Munich, Germany

INTERNSHIP EXPERIENCES

AI Large Model Engineer BOSCH (China)	Apr 2025 – Jul 2025
- Optimized code deployment workflow. Developed a unit test code generator by fine-tuning Bosch's internal LLM with LoRA on 4,000+ GTest code pairs, improving test coverage by 12% (79%→91%) and enabling 300+ daily API calls. - Built an end-to-end semantic understanding system for autonomous driving requirements using QLoRA, converting client language into standardized technical documents and reducing development cycle time by 30%+.	
Software Development Engineer OPPO	Aug 2024 - Oct 2024
- Developed an automated project progress alert bot using Python, implementing multithreaded web crawlers to monitor site parameters, detect anomalies, and trigger real-time alerts via TeamTalk API, achieving 100% anomaly interception. - Contributed to OPPO App Store release, focusing on Android UI iteration and function optimization. Wrote and executed 23 Kotlin test cases using Android Studio, intercepting 4 P0-level critical bugs and ensuring on-time delivery.	
Machine Learning Intern CITIC Securities	Feb 2024 - Mar 2024
- Improved stock trend prediction by building an XGBoost model incorporating MA(30,60,120) and MACD(12,26,9) indicators; performed nonlinear modeling on CSI 300 historical data, achieving 70%+ forecast accuracy. - Used XGBoost and SHAP to extract key stock price drivers, apply dynamic weighting, and boost returns.	
ML Intern Hong Kong Applied Science and Technology Research Institute (ASTRI)	Aug 2023 – Sept 2023
- Developed a hybrid Polynomial Regression + Random Forest model to optimize silicon wafer performance by modeling temperature-performance relationships, and achieving fitting performance curves. - Built a battery RUL prediction algorithm using a Regression model, providing maintenance planning for companies.	

RESEARCH EXPERIENCES

Instrumental Music Search Based on Playing Characteristics under Prof. Clement Leung	Shenzhen, China
Independent Researcher, School of Science and Engineering (SSE), CUHK(SZ)	Sept 2024 – May 2025
- Developed a CNN-based model integrating MFCC and spectrogram, achieving 92% accuracy in musical recognition. - Designed fusion architecture combining Mel-Spectrogram and temporal features; incorporated dilated convolutions and attention mechanisms to enhance long-sequence modeling. Related work published in a research paper (1).	
Safe Reinforcement Learning for Modular Robots under Prof. Matthias Althoff	Munich, Germany
Research Group Member, Leibniz Supercomputing Centre (LRZ), TUM	Apr 2024 - Aug 2024
- Developed synchronization between robotic simulations and real-world operations to reduce discrepancies - Optimized critic-actor algorithms to ensure safe operations and minimize collisions [poster][report]	
Adaptive ML Algorithm Design for Marine Robots under Prof. Yangsheng Xu	Shenzhen, China
Research Assistant, Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)	Oct 2022 - Feb 2024
- Applied Kalman filter and polynomial regression to reduce noise and improve sailboat positioning accuracy by 83%. - Developed an autoregressive model-based adaptive algorithm to enable dynamic sailboat navigation; published paper 2.	

PUBLICATIONS

- 1) X. Y. Zhu, C. Leung (2025). CNNs in Musical Performance and Arrangement: Recognizing and Managing Bowed Instrument Techniques Across Cultures, AIMEDIA, Venice, Italy.
- 2) X. Y. Zhu, C. Liang, H. H. Qian* (2023). [Design and Implementation of a Novel Adaptive Multihull Sailboat with Lifiable Side Hulls](#), 2023 IEEE International Conference on Robotics and Biomimetics (ROBIO), Koh Samui, Thailand.